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# Interview questions on Pressure Measurement Instrumentation Tools

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9-12 minutes

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### 1.what are the application advantages of a dead weight tester and a gauge comparator?

Dead weight tester: Generally it is used for calibrating a test gauge. It works on the hydraulic principle, where a test gauge is compared with the standard dead weights.

Calibration procedure is laborious.

Gauge comparator: It is used for calibrating a pressure gauge against a test gauge. Calibration procedure is simple and quicker.

### 2.How would you select a pressure gauge for a process?

While selecting a pressure gauge for a process, consider the following points:

1.Characteristic of the process (corrosive or non-corrosive)

2.Operating pressure

3.Maximum process pressure

The gauge range should be a minimum of twice the process operating pressure.

**3. Write the types of Bourdon tubes? Explain the purpose of different Bourdon tubes.**

The following are the commonly used Bourdon tubes used in industries:

– Spiral –Low range 10 –100kpa

– C type –Medium range 100 – 5000kpa

– Helical – High range 5000 – 20000kpa

(refer to the manufacturers manuals for correct parameters)

**4.What is function of a hair –spring in a pressure gauge?**

Hair – spring in a pressure gage eliminates the Hysterisis (backlash/angularity) error caused in the quadrant (gear and pinion mechanism) and the weight of the pointer.

**5.Name a few pressure switch manufacturing companies**.

Barksdale, United electric, CCS…etc.

**6.What is the output of a pressure switch?**

Output of an electric switch is a contact- open or close.

Out put of a pneumatic switch is a port operation- open or close, vent

**7.How would you select a pressure switch for 500kpa Hi operatin?**

The following points are to be considered wile selecting a pressure switch for any process operation:

– The process operating pressure

– The maximum process pressure

– The process pipeline vibration

– The maximum working pressure of the pressure switch should be times greater then the maximum operating pressure

– The micro switch contact rating

– Process connection

– Local or remote mounting …etc.

**8.Expain the contact selection on high and low pressure alarm switches? And explain why?**

On a high pressure switch, the wiring is terminated on the “common” and the “normally close” contact terminals.

On a low pressure switch, the wiring is terminated on the “common” and the “normally open” contact terminals.

This type of contact termination is done to achieve a close contact from the switch during a normal (healthy) process condition which is a fail safe method.

**9.What is to be done, if a transmitter gives a maximum output , where the transmitter range (jumper) is already in maximum selection?**

Replace the capsule (sensor) for a higher range in the transmitter.

**10.Explain how to carryout a field zero check on a pressure transmitter?**

– If the transmitter signal is used for controlling, then put the controller on manual.

– Connect a digital multimeter in the current range in series with the transmitter output.

– Isolate the process valve

– Isolate the instrument isolation valve.

– Open the equalising valve

– Open the bleed valve and de-pressurise the transmitter.

– Check for 4.00mA output signal, if not, adjust the zero screw.

**11.What-test equipment is required to calibrate a pressure transmitter in the field?**

If it is a low range transmitter, then precision pneumatic test equipment like “wallace and tiernier” and pneumatic pump (or air supply) may be used.

If it is a high range transmitter, then a hydraulic pump and a test gauge is to be used.

**12.What are pressure measuring/feeding test equipment?**

The following are a few commonly used test equipment to measure and feed pressure: Tradinco, Wallance & tiernier, Ralston, Gauge comparator, Dead weight tester…etc.

**13.Write the formula for calculating a static head in kpa?**

Static Head = pgh

p= Density of the liquid

g= Gravity

h= height of the liquid column

**14.What are the simple methods for measuring level?**

Gauge glass, Pressure Gauge – scale graduated in % of level, Rope and weight , Dip Tape…etc

### 15.How to convert a pressure gauge into a level gauge?

Calculate the static head in kpa using the formula “pgh”. Select a pressure gauge and calibrate it for the calculated static head. Graduate the pressure gauge scale in terms of %of level.

### 16.What is the density of crude oil?

The density of crude oil is approximately 0.8.

### 17.What is an “interface dip tape”? Where it used?

Interface dip tape is an instrument used for measuring the total and the interface liquid level in vessel.

For example: It is used in oil storage tanks to measure the total liquid level and water level.

### 18.Explain how a Leveltrol works? Name the parts of a pneumatic Leveltrol.

A Leveltrol in and instrument used for measuring the liquid level between two known points. The Leveltrol works on the buoyancy principal. Leveltrol has a float, which submerses proportionately with liquid level raise in the float chamber. The amount of submersion of the displacer depends on the liquid density, which produces a torque. The amount of torque produced in measured in terms of % of level.

### Parts of Leveltrol:

Float chamber, Float, Torque lever, Knife edge, Feedback Bellows, Air Relay, Restriction, Flapper, Nozzle, Feedback link, Density range, Action change lever, HP and LP flange,

### 19.What is a static head level transmitter?

A static head level transmitter is used for measuring the total liquid

level in the tank.

**20.What are the application limits of a Leveltrol and a static head level transmitter?**

A Leveltrol can measure only for a short and fixed height of level. A Leveltrol has a high gain output. In can be used for liquid level as well for the interface level measurement.

Static head transmitter is used for measuring the total liquid level in the tank. The transmitter output is linear.

**21.What is the operating principle of a Magnetrol level switch?**

The Magnetrol level switch works on the principle of Buoyancy force.

**22.What are the points to consider while selecting a level switch for a particular process?**

While selecting a level switch, the following points are to be considered:

– The characteristic of process (corrosive or non – corrosive)

– The process pressure

– The liquid density

– The flange ratings

– Proof pressure of the switch

– The micro switch contact rating

**23.What type of level measuring instrument is suitable for closed tank?**

If the level measurement is required for a fixed and small height, a Leveltrol is more suitable.

If it is for a total height, then either a static head or a differential pressure transmitter with its HP leg connected to the bottom of the liquid level and the LP leg connected to the top of the tank to gas phase.

### 24.What is Zero suppression and Zero elevation in level measurement?

**Zero suppression:** when a static head transmitter is installed below the zero liquid level, the transmitter gets a +ve error in the level measurement. This error is corrected by a zero suppression kit.

**Zero elevation:** when a static head transmitter is installed above the zero liquid level, the transmitter gets a –ve error in the level measurement. The error is corrected by a zero elevation kit.

### 25.What is an interface in level measurement?

An interface is the separation point between the two de-missible liquids levels In a vessel. This condition arises when the liquid does not mix due to its chemical composition and difference in their density.

### 26.How to calibrate a leveltrol for an interface level measurement?

Fill the leveltrol chamber 100% with the lower density liquid and adjust its zero for 4.00mA output.

Drain the liquid and fill the leveltrol chamber 100% with the higher density liquid and adjust its span for 20.00 mA output.

The transmitter on line measures the percentage of higher density liquid in the lower density liquid at a known height.

### 27.How to calibrate a static head level transmitter for an interface measurement?

Static head level transmitter is not commonly used for measuring the interface level. But the following procedure may be used for calibrating it to measure the interface level in a tank.

Fill the vessel 100% with the lower density liquid and adjusts its zero for 4.00 mA output. Drain the liquid and fill with the vessel 100% with the higher density liquid and adjust its span for 20.00mA output. 100% transmitter level is to be continuously maintained.

The transmitter on line measures the percentage of higher density liquid in the lower density liquid in a know height.

### 28.Why is a leveltrol more suitable tan a static head level transmitter on a separator?

Leveltrol is more accurate in measuring the small height of liquid level. The process pressure change does not affect the leveltrols performance. Leveltrol offers a high gain output verses the change in the liquid level.

### 29.Why is static head level transmitter more suitable than a leveltrol on a surge tank?

A leveltrol has a limitation in measuring the maximum height of liquid. Considering huge of the surge tank, a static head level transmitter is more suitable.

### 30. Why is a gap controller often used in controlling a level in a high pressure vessel?

Gap controllers output changes from minimum to maximum and visa versa when the process measurement deviates the setpoint by the set gap. Hence the final control element i.e. control valve acts

like an on-off valve. This helps in minimizing the trim erosion due to a high DP across the control valve while partially open.

### 31. How to use a DP (differential pressure) transmitter for level measurement?

A differential pressure transmitter gives a linear output for the differential pressure measured across its HP and LP chambers.

Connect the transmitter's HP leg to the bottom of the tank and its LP leg to the top of the tank. The transmitter will read the liquid level accurately irrespective of the change in process pressure above the liquid surface.